

No. of Printed Pages : 06

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43501

Roll
No.

B. Tech. Examination, 2026

(Even Semester)

ENGINEERING MECHANICS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$
- (a) Define law of transmissibility.
 - (b) Write equilibrium equation for concurrent force system.
 - (c) Define simply-supported beam (SSB).
 - (d) Define angle of friction.
 - (e) Define perfect truss.
 - (f) Define centroid and moment of inertia (MoI).

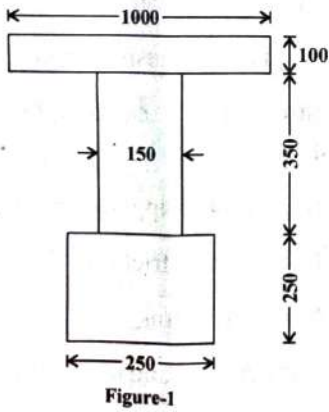
[P. T. O.]

- (g) Define work-energy principal.
- (h) Define Young's modulus of elasticity.

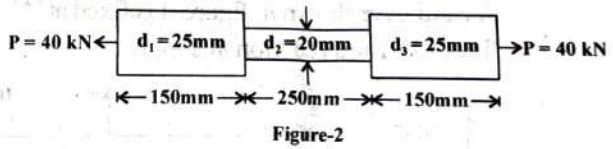
SECTION – B

2. Attempt any two parts of the following : 2×6=12

- (a) The resultant of two forces. One of them is double to another is 260N. If the direction of large force is reversed and other remain unchanged. Find the forces. $R_2 = 180\text{ N}$
- (b) Define beam, types of beam, types of loading.
- (c) The cross-section of pre-stressed concrete beam is shown in figure-1. Calculate the moment of inertia of this section about the centroidal axes parallel to and perpendicular to top edge :



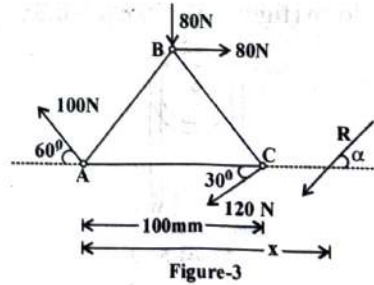
- (d) A bar is shown in figure-2 is tested in universal testing machine. It is observed that load of 40kn bar gives total extension of 0.280 mm. Find Young's modulus of the material :



SECTION – C

Note :- Attempt all questions. Attempt any two parts from each questions. 8×5=40

- 3. (a) What do you mean by term free body diagram? Explain with example.
- (b) Find the resultant of the force system in figure-3 acting on a lamina of equilateral triangular shape. Also find the distance "x" from point A :



[P. T. O.]

(c) Explain the following :

- (i) Law of parallelogram of force.
- (ii) Lamis' theorem.

4. (a) A cantilever shown in figure-4 is fixed at 'A' and free at B. Find reaction at support :

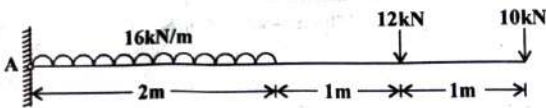


Figure-4

(b) Define and explain the following :

- (i) Law of dry friction.
- (ii) Coefficient of friction.
- (iii) Angle of friction.

- (c) Determine the maximum weight that can be lowered by a person who can exert at 300 N pull on rope if the rope is wrapped $2\frac{1}{2}$ turn round the drum (figure-5). Take $\mu = 0.3$:

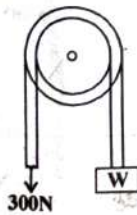


Figure-5

5. (a) State atleast three assumption in the analysis of truss.
- (b) Find forces in all members of loaded truss figure-6 :

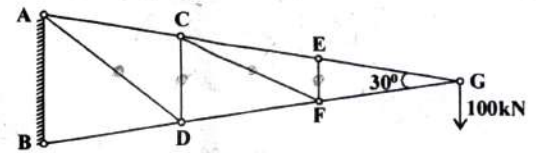


Figure-6

- (c) From first principal determine the centroid of a TRIANGLE from base.
6. (a) A motorist is travelling with a speed of 70 kmph suddenly applies breaks and halts after skidding 50 m. Determine :
- (i) Time required to stop the car.
 - (ii) The coefficient of friction between tyres and the road.
- (b) Draw stress-strain curve for mild steel and explain the salient points on it.
- (c) The motion of particle defined as :

$$s = 16t + 4t^2 - 3t^3$$

where s in meter and t , in second. Determine :

- (i) Displacement, velocity and acceleration 2 sec. after start.
- (ii) Displacement and acceleration when velocity is zero.
